AI-Assisted Epilepsy Detection using EEG Signals

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Outline

Problem and Motivation

- Epilepsy affects **50 million** people worldwide; **30%** have drug-resistant epilepsy requiring long-term EEG monitoring
- Clinical diagnosis relies on **visual inspection** of EEG by neurologists
  - Hours of multi-channel recordings per patient — extremely time-consuming
  - High **inter-rater variability** — subjective interpretation leads to inconsistent diagnoses
  - Shortage of trained neurologists, especially in developing regions
- Automated EEG analysis can provide **objective**, **consistent**, and **scalable** diagnostic support
- Evolution of automated methods:
  - Traditional ML: hand-crafted features (wavelet, entropy) + SVM / Random Forest
  - Deep learning: CNN (spatial) → RNN/LSTM (temporal) → **hybrid architectures**
  - Hybrid models combine strengths of multiple components, but lack rigorous evaluation

Research Gaps and Objectives

Three critical gaps in existing literature:

1. **Data leakage** — Most studies (e.g., Thara 2019) split at *segment level*; overlapping sliding windows from the same recording appear in both train and test → artificially inflated accuracy
2. **Lack of ablation studies** — Hybrid models are proposed as a whole, rarely decomposed to verify each component's contribution (Shoeibi 2021)
3. **Limited multi-class evaluation** — Binary (normal vs seizure) is dominant; clinically important *interictal* states are often ignored

Project objectives:

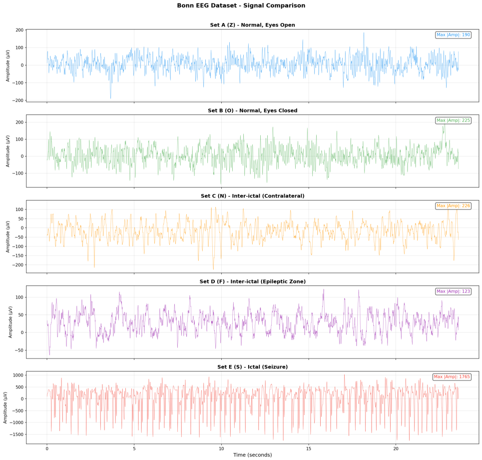
1. Implement **recording-level** 10-Fold CV to eliminate data leakage entirely
2. Design a lightweight hybrid **CNN + Bi-LSTM + Self-Attention** architecture (~500K params)
3. Conduct systematic **ablation**: 3 baselines (Pure CNN, Pure BiLSTM, SVM+WPD)
4. Evaluate across **3 tasks** of increasing difficulty: binary, three-class, five-class
5. Visualise attention weights to enhance **clinical interpretability**

Dataset — University of Bonn EEG

- **5 subsets**, 100 single-channel recordings each (500 total)
- 173.61 Hz sampling, 23.6 s/recording, 4097 points
- Bandpass filtered 0.53–40 Hz during acquisition

Set	Subject	State	EEG Type
Z	Healthy	Eyes open	Surface
O	Healthy	Eyes closed	Surface
N	Epileptic	Interictal, contralateral	Intracranial
F	Epileptic	Interictal, epileptogenic	Intracranial
S	Epileptic	**Seizure (ictal)**	Intracranial

Task	Classes	Rec.
Binary	Normal (Z+O) vs Seizure (S)	300
Three-class	Normal vs Interictal vs Seizure	500



EEG waveform comparison across five subsets:

Normal (Z, O) shows regular low-amplitude activity;

Data Leakage: Problem and Solution

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The Problem — Commonly Overlooked in Literature

Many studies (e.g., Thara 2019) split EEG *segments* randomly. With 50% overlapping windows, adjacent segments from the **same recording** share up to half their data points → model memorises patterns → **artificially inflated accuracy**.

Our Solution — Recording-Level 10-Fold Cross-Validation

- Split at **recording level**: 500 recordings assigned to folds **before** any windowing
- Each fold: 450 train / 50 validate — **zero overlap** between sets
- Sliding windows (1024 pts, 50% overlap) generated **independently within each set**
- StandardScaler **fit on training windows only**, then applied to validation

Discovered during development when initial accuracy seemed suspiciously high. After fix, results remain strong and consistent with peer-reviewed benchmarks.

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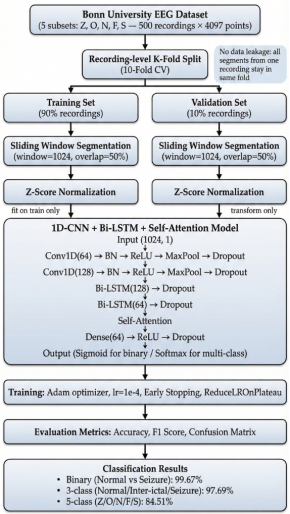


1. Recording-level Split

500 recordings → 10 folds.

All windows from one recording stay in the same fold.

2. Sliding Window

1024 points (∼5.9 s), 50% overlap.

Applied independently per fold.

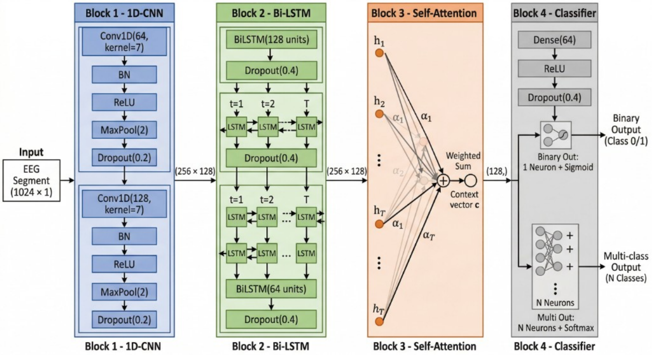
3. Z-Score Normalisation

Global mean/std computed on **training set only**; applied to validation set.

4. No Extra Filtering

Bonn dataset already bandpass filtered at 0.53–40 Hz during acquisition.

Proposed Hybrid Model Architecture



1D-CNN
2 Conv layers (64, 128)
Local morphological features
(spike shape, sharp waves)

Bi-LSTM
2 layers (128, 64 units)
Bidirectional temporal
dependency modelling

Self-Attention
$\mathrm{softmax}(QK^\top/\sqrt{d_k})\,V$
Focus on clinically
relevant time steps

Output: Dense(64) → ReLU → Dropout → Softmax | Total: ∼**500K** parameters

Ablation Study Design

Model	CNN	BiLSTM	Attention	Purpose
Hybrid (ours)	✓	✓	✓	Full proposed model
Pure CNN	✓			Spatial features only
Pure BiLSTM+Attn		✓	✓	Temporal features only
SVM + WPD		(traditional ML)		Non-DL baseline

- **Controlled comparison**: all DL models share identical preprocessing, data splits, and 10-Fold CV protocol — only the architecture varies
- **SVM baseline**: Wavelet Packet Decomposition (3-level, 'db4') extracts energy and Shannon entropy features per subband; RBF kernel SVM classifier
- **Ablation logic**: removing CNN isolates temporal modelling (BiLSTM); removing BiLSTM+Attention isolates spatial features (CNN); SVM tests traditional ML upper bound
- **Training**: Adam optimiser, lr$=10^{-4}$, early stopping (patience=10), ReduceLROnPlateau; trained on RTX 3070 Ti, 50 epochs per fold

Results — Binary and Three-class Classification

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Binary: Normal (Z+O) vs Seizure (S)

Model	Acc (%)	F1
SVM+WPD	99.67±1.00	99.66
Pure CNN	98.86±1.91	98.84
BiLSTM+Attn	99.48±1.29	99.47
Hybrid	**99.67**±0.85	**99.66**

Binary task is inherently simple due to distinct morphological differences between normal and seizure signals. All models >98%.

Three-class: Normal / Interictal / Seizure

Model	Acc (%)	F1
SVM+WPD	94.20±2.44	94.19
Pure CNN	97.14±1.84	97.11
BiLSTM+Attn	92.80±5.99	92.74
Hybrid	**97.69**±2.17	**97.68**

Introducing interictal class increases difficulty. BiLSTM alone drops to 92.80% with high variance (±5.99), while Hybrid remains stable at 97.69%.

Results — Five-class Classification

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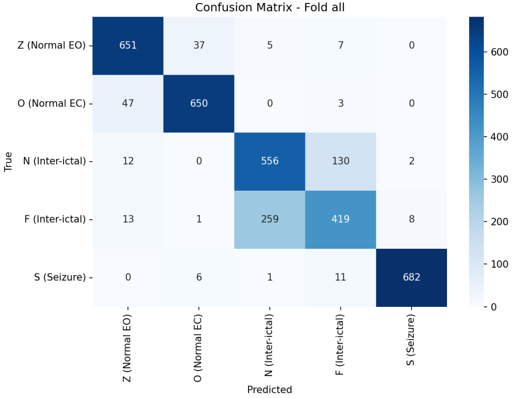
Model	Acc (%)	F1
SVM+WPD	78.80±5.31	78.39
Pure CNN	78.86±3.77	75.95
BiLSTM+Attn	73.09±8.37	71.92
Hybrid	**84.51**±3.62	**84.28**

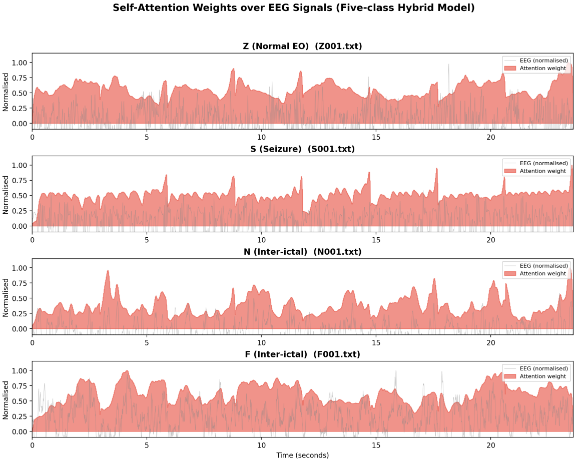
- Hybrid leads CNN by **+5.65%**
- Hybrid leads BiLSTM by **+11.42%**
- CNN captures local shape but misses long-range context; BiLSTM lacks spatial precision
- **Synergy** of all three components is most evident in this hardest task



Five-class confusion matrix (Hybrid model).
Main errors: N↔F mutual confusion — both are interictal recordings with similar morphology.

Z and S classes achieve near-perfect classification.

Attention Weight Visualisation



Class-specific attention patterns:

- **Seizure (S)**: high weights on *sharp waves* and *high-frequency bursts* — hallmark of ictal activity
- **Normal (Z, O)**: weights distributed *uniformly* — no dominant abnormal region
- **Interictal (N, F)**: moderate peaks on subtle *spike-and-wave* complexes

Clinical significance: Attention aligns with features neurologists look for during visual EEG review. This “white-box” property supports clinical trust in AI-assisted diagnosis.

Statistical Significance & Literature Comparison

Paired *t*-test on 10 fold accuracies:

Comparison	t	p	Sig.
Hybrid vs CNN	4.98	0.0008	**
Hybrid vs BiLSTM	4.08	0.0028	**
Hybrid vs SVM	2.73	0.0230	*

All $p < 0.05$ → Hybrid's advantage is **statistically significant**, not due to random fold variation.

* $p < 0.05$ ** $p < 0.01$

Comparison with published studies:

Study	Method	Task	Acc
Ullah 2018	P-1D-CNN	Binary	99.1%
Thara 2019	Stacked BiLSTM	Binary	99.0%
Acharya 2018	13-layer CNN	3-class	88.7%
Huang 2025	STFFDA	5-class	67.2%
Ours	**Hybrid**	**5-class**	**84.5%**

+17.3pp over Huang et al. on same Bonn 5-class 10-Fold CV.
Thara 2019 used segment-level split (potential leakage); our recording-level split ensures **fair** comparison.

Limitations and Further Work

Current Limitations:

- **Single dataset**: Bonn has only 500 recordings from 10 subjects; limited generalisability
- **Single-channel EEG**: cannot exploit spatial relationships across electrode montages
- **Class imbalance**: Binary 2:1 ratio; no explicit handling (pos_weight, oversampling)
- **Offline system**: processes pre-recorded segments; not designed for real-time streaming

Proposed Further Work:

- **Multi-channel datasets**: CHB-MIT (24 patients, 916h) and TUH EEG Corpus (20K+ records)
- **Hyperparameter search**: Bayesian optimisation for lr, hidden size, window length
- **Transfer learning**: pre-train on Bonn, fine-tune on larger clinical datasets
- **Real-time inference**: ONNX/TensorRT on wearable EEG devices
- **Patient-adaptive**: personalised fine-tuning for individual EEG patterns

Key Contributions of This Project:

1. **Leakage-free evaluation** — Recording-level 10-Fold CV prevents the data leakage problem found in many prior studies, ensuring rigorous and reproducible benchmarking
2. **Hybrid architecture excels on hard tasks** — 1D-CNN + Bi-LSTM + Self-Attention achieves **84.51%** five-class accuracy, statistically outperforming all baselines ($p < 0.05$)
3. **Systematic ablation** — Each component (CNN, BiLSTM, Attention) contributes measurably; single-architecture models lose 5–11% on five-class
4. **Interpretable attention** — Weight visualisation aligns with clinical EEG reading patterns, supporting model trustworthiness for clinical adoption

Thank You

Questions?